DRIVER'S INTERFACE DESIGN FOR A FORMULA STUDENT VEHICLE

Formula Student FEUP 03

Department of Chassis

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2025-2027

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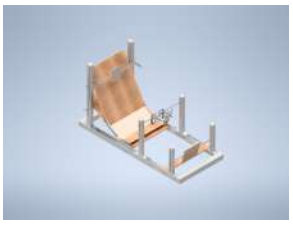


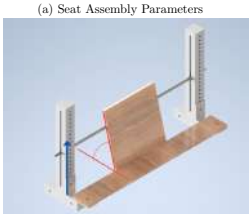
Figure 1.1: FS FEUP 02 Ergonomic JIG Isometric View



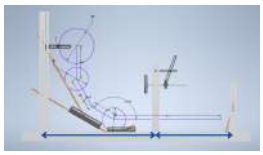
(a) Seat Assembly Parameters



(b) Steering Wheel Assembly Parameters



(c) Pedal Box Assembly Parameters



(d) Global Assembly Parameters

Figure 1.2: FS FEUP 02 Ergonomic JIG Parameters

The variable parameters were the following:

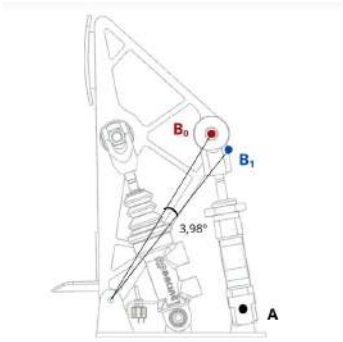


Figure 2.9: Points considered in the pneumatic actuators travel calculations

The points shown below were defined such that A represents the lower mounting point and B_0 represents the upper mounting point in the resting position. Using this geometric setup, the objective is to determine the horizontal coordinate x of the lower mounting point that results in a compression of 8 mm during the full pedal travel.

$$A = (x,\ -8)$$

$$B_0 = (86.46,\ 147.72)$$

The coordinates of point B_1 must be calculated as a rotation of B_0:

$$B_1 = \begin{pmatrix} \cos\Delta\theta & -\sin\Delta\theta \\ \sin\Delta\theta & \cos\Delta\theta \end{pmatrix} B_0$$

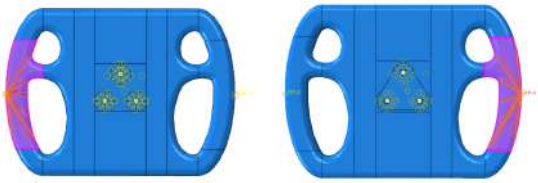
The following equations allow us to compute the actuator length in both the initial position and the final position after pedal rotation, given the defined compression stroke.

$$L_0(x) = \sqrt{(x - B_{0x})^2 + (-8 - B_{0y})^2}$$

$$L_1(x) = \sqrt{(x - B_{1x})^2 + (-8 - B_{1y})^2}$$

The value of x can be determined through the following equation:

$$L_0(x) - L_1(x) = 8 \text{ mm} \Leftrightarrow \sqrt{(x - B_{0x})^2 + (-8 - B_{0y})^2} - \sqrt{(x - B_{1x})^2 + (-8 - B_{1y})^2} = 8$$



(a) (b)

Figure 2.60: Coupling constraints.

A Tie constraint was applied to the surfaces of the steering wheel and the faces of the insert that will be in contact (Figure 2.61).

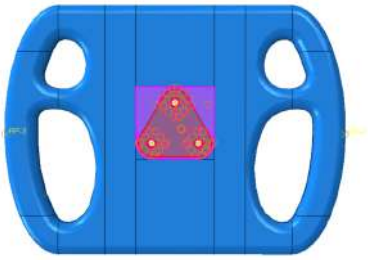
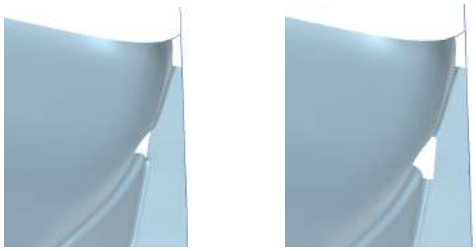


Figure 2.61: Tie Constraint.

Loads and Boundary Conditions

For boundary conditions the walls of the three quick release holes were pinned. Three different loads were applied to the reference points associated with the grip surfaces:

- Two forces of 740 N with opposite directions were applied to the reference points to cause a moment of 162.8 N.m. This value comes from the ergonomic study performed in Labiomep, where the maximum torque registered in the study was 81.4 N.m.
- Two longitudinal forces of 660 N were applied in the same direction. This value comes from literature.



(a) (b)



(c)

Figure 2.92: Third blend.

There one other imperfection that needs to be altered which is the end of the first blend (Figure 2.93a). So, that portion was deleted and then covered using Fill Surface, followed by an Edge Blend (Figure 2.93b).